



AUTOMATED IDENTITY MANAGEMENT FOR EMBEDDED COMPONENTS

MARCH 2026

FLORIAN HANDKE

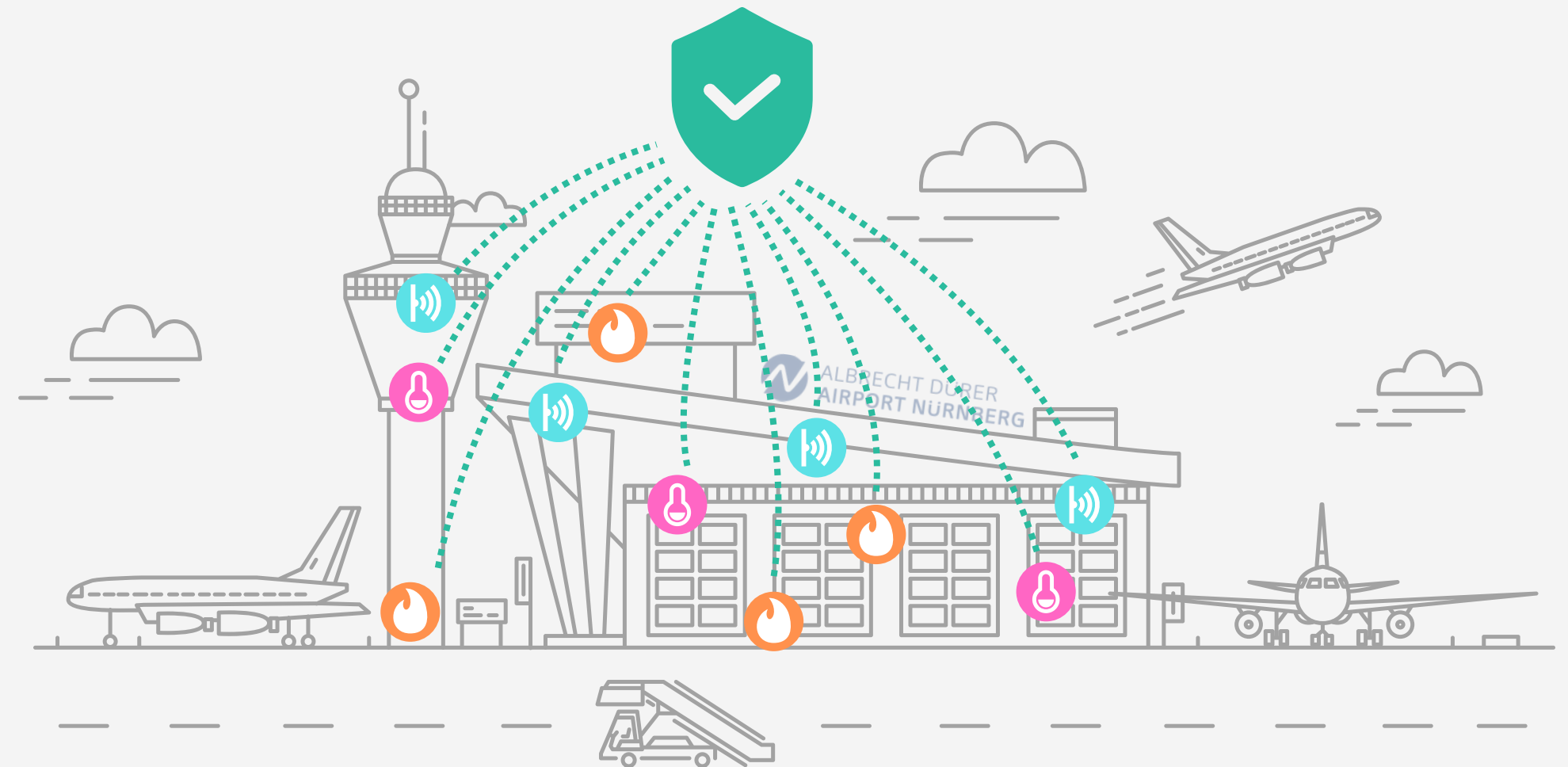
THE CHALLENGE

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How can a device securely and automatically join a local trust domain without cloud services, manual steps, or vendor lock-in?

When a new device is commissioned, it should:

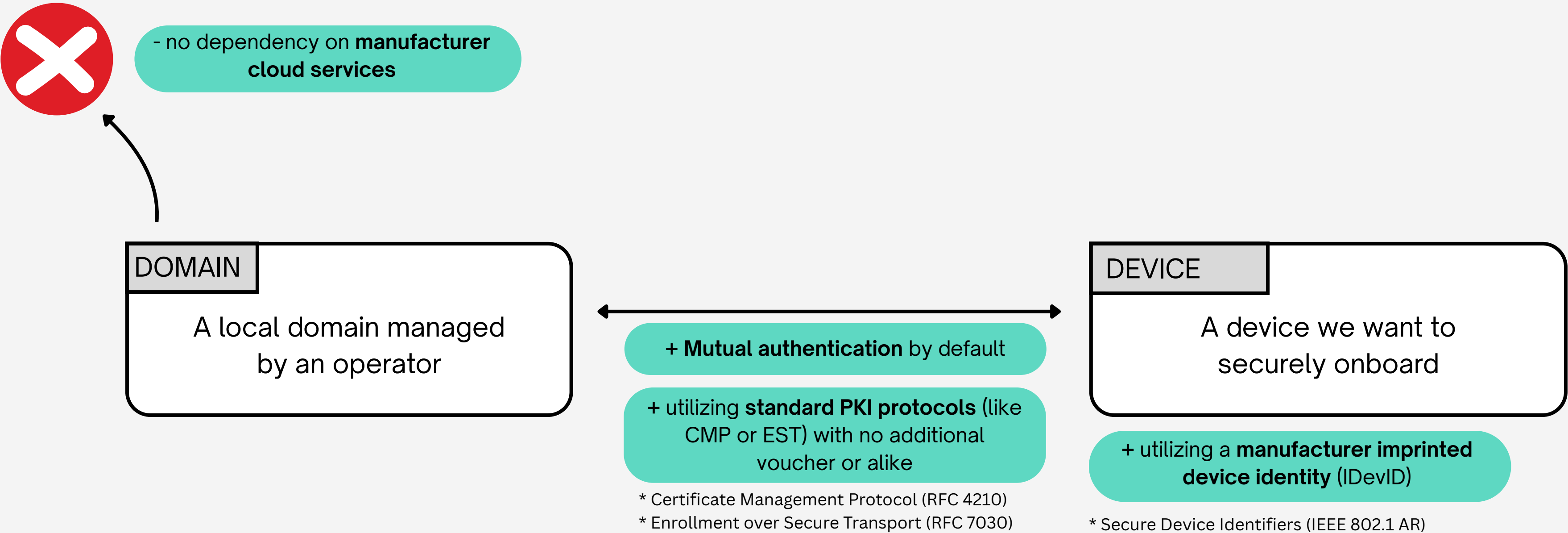
- Automatically **join the correct security domain**
- Prove its **authenticity**
- Receive **operational credentials**
- Establish **trusted communication**
- Require **zero manual** (certificate) **handling**



GOAL & CONSTRAINTS

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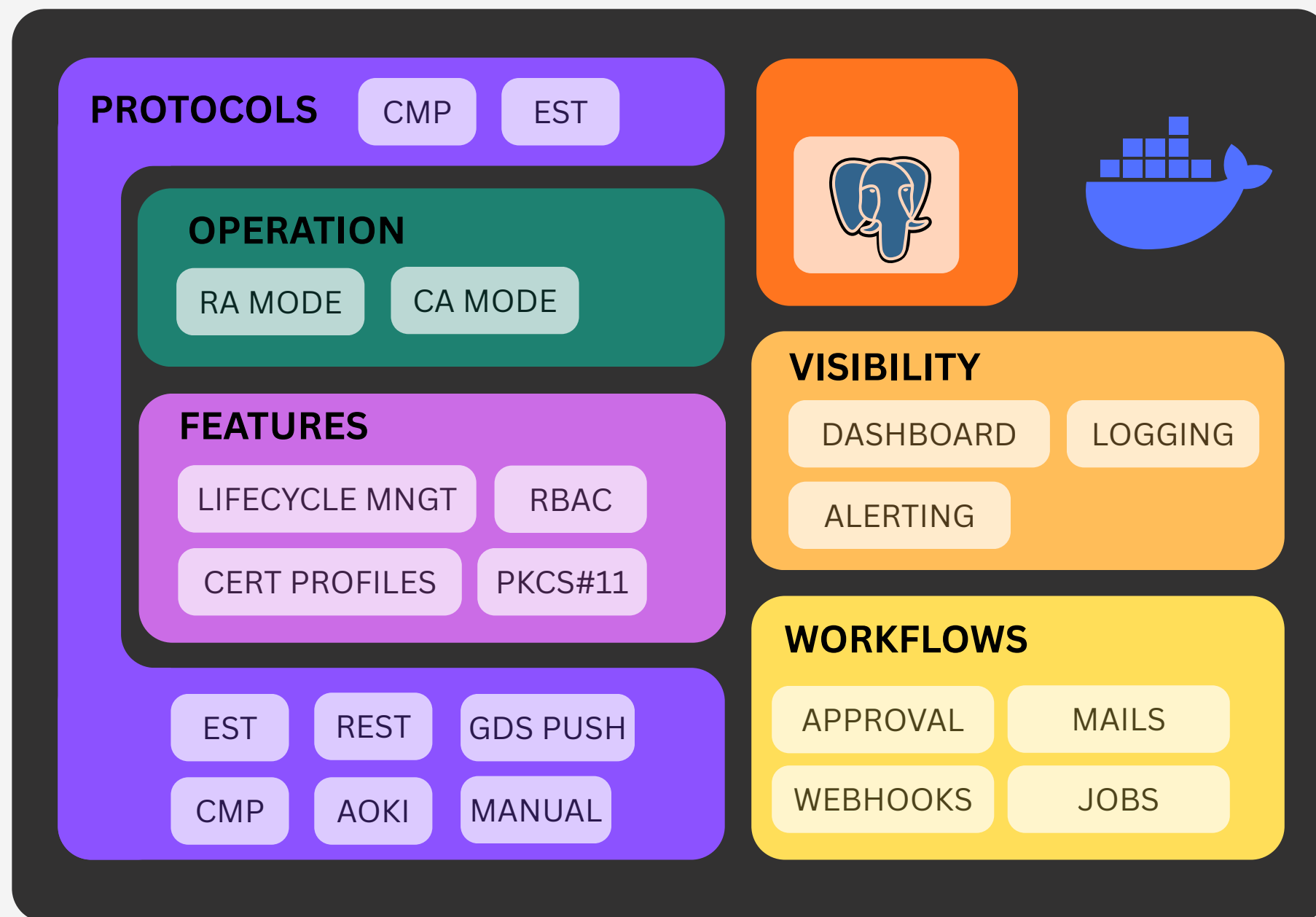
Trust establishment must work locally, autonomously, and without third-party runtime dependencies.



TRUSTPOINT

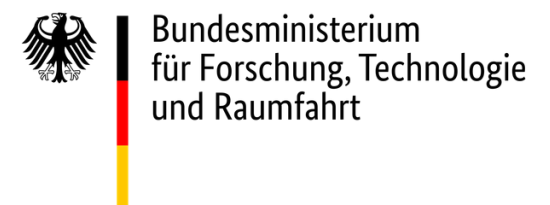
Open-Source Platform for Managing Digital Machine Identities in Industrial Environments

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- Open Source (MIT License)
- Fully Offline-Capable
- Designed for Device Manufacturers, System Integrators, and Operators
- Applicable to Industry, Building Automation, and Energy Networks
- Developed by an Industry & Research Consortium

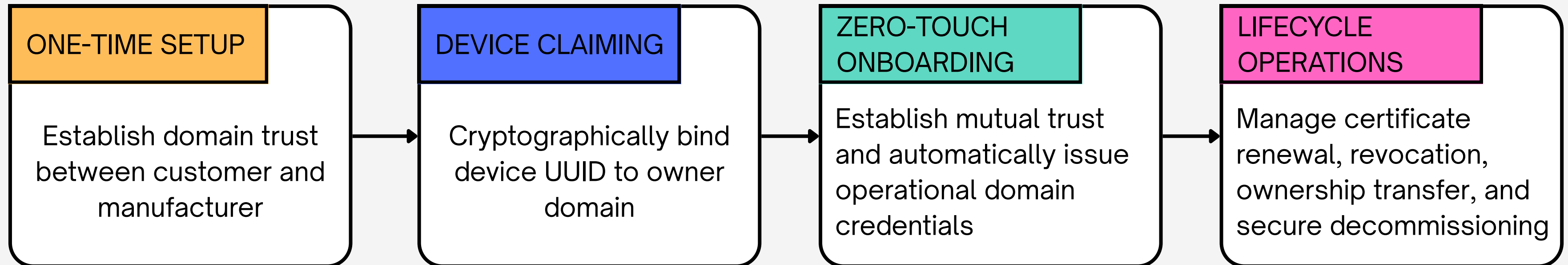
Funded by:



AOKI WORKFLOW

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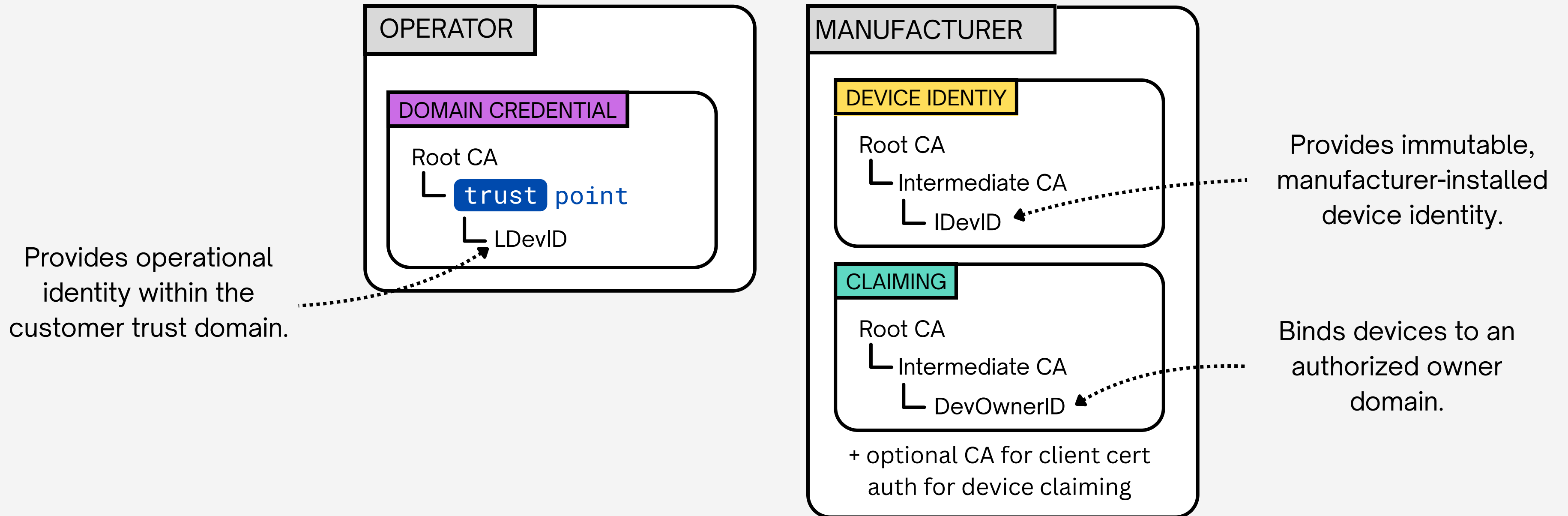
A Structured Four-Phase Model for Secure
Domain Registration, Ownership Binding, Zero-
Touch Onboarding, and Long-Term Identity
Lifecycle Management



PKI SETUP

Separation of Device Identity, Ownership, and Operational Domain Trust

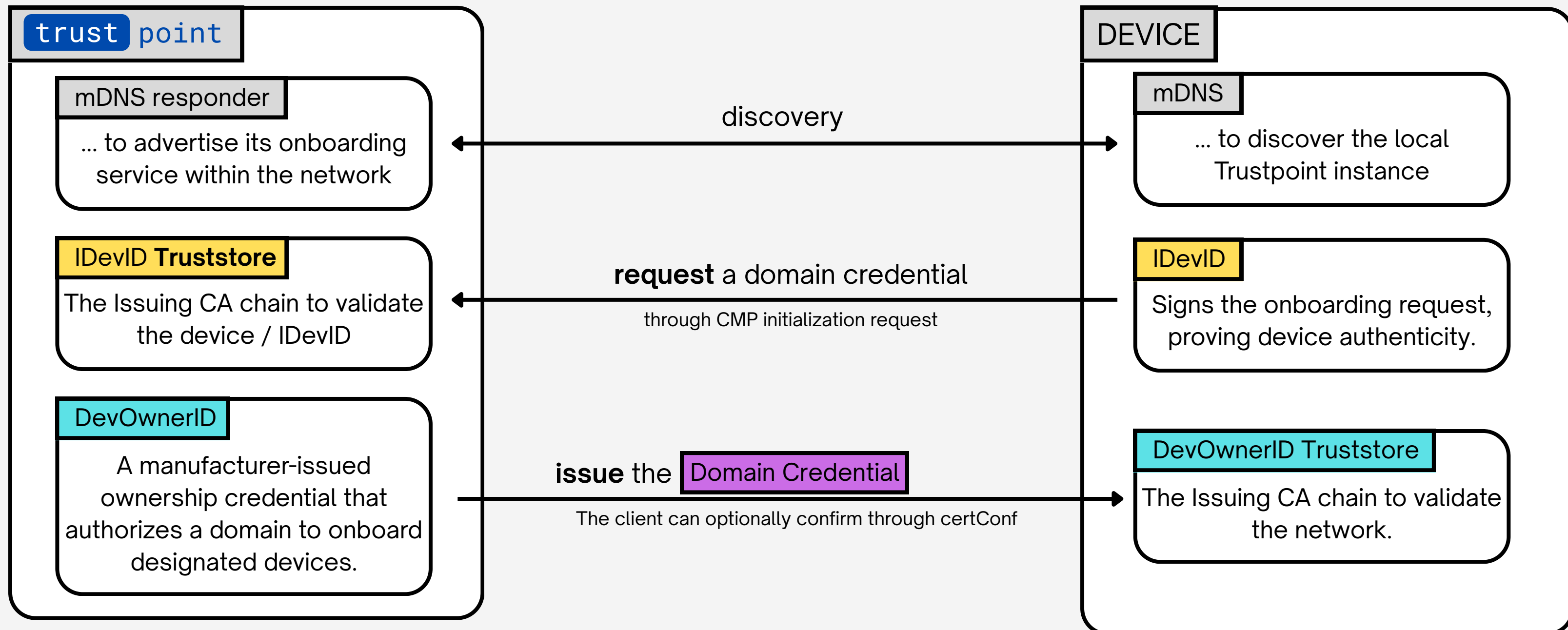
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ZERO TOUCH ONBOARDING

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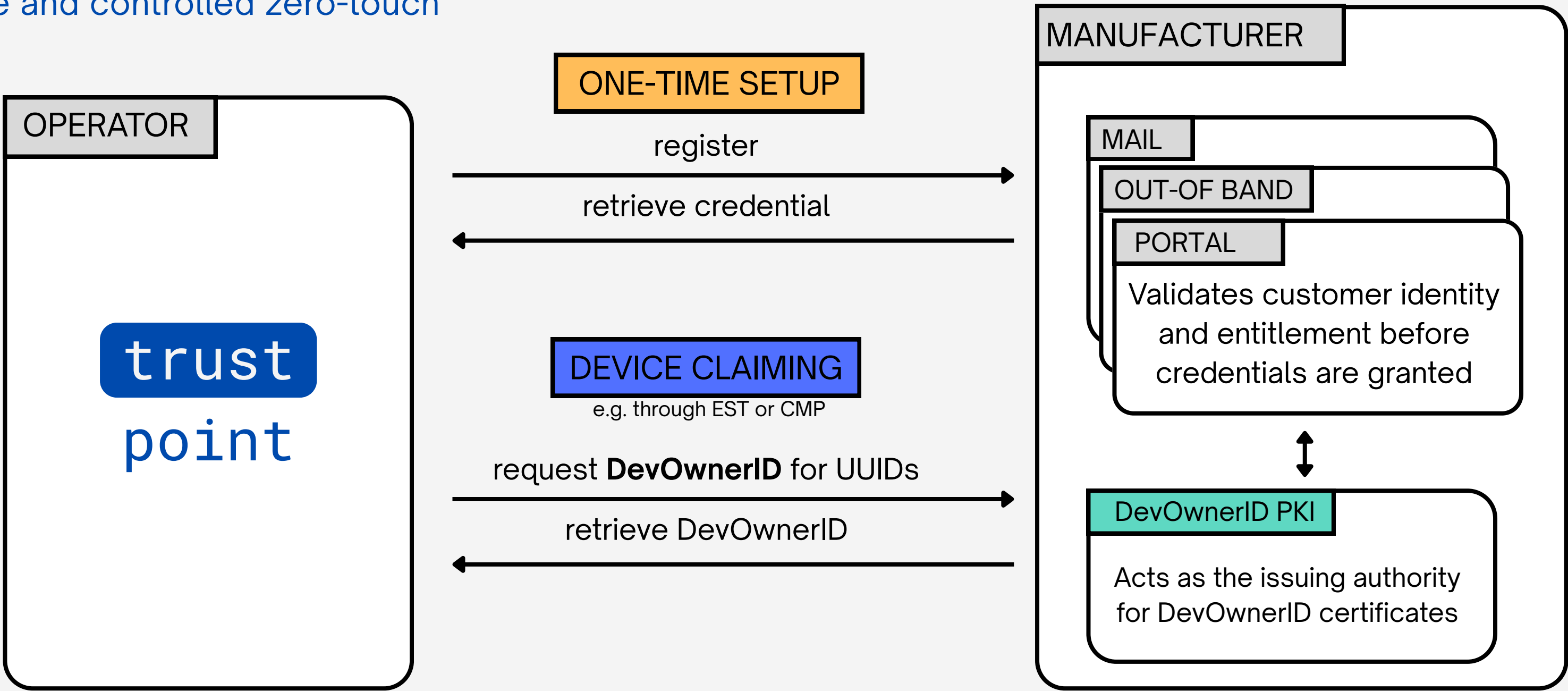
Securely bootstrapping devices into the owner domain through automated mutual authentication and credential provisioning



SETUP & DEVICE CLAIMING

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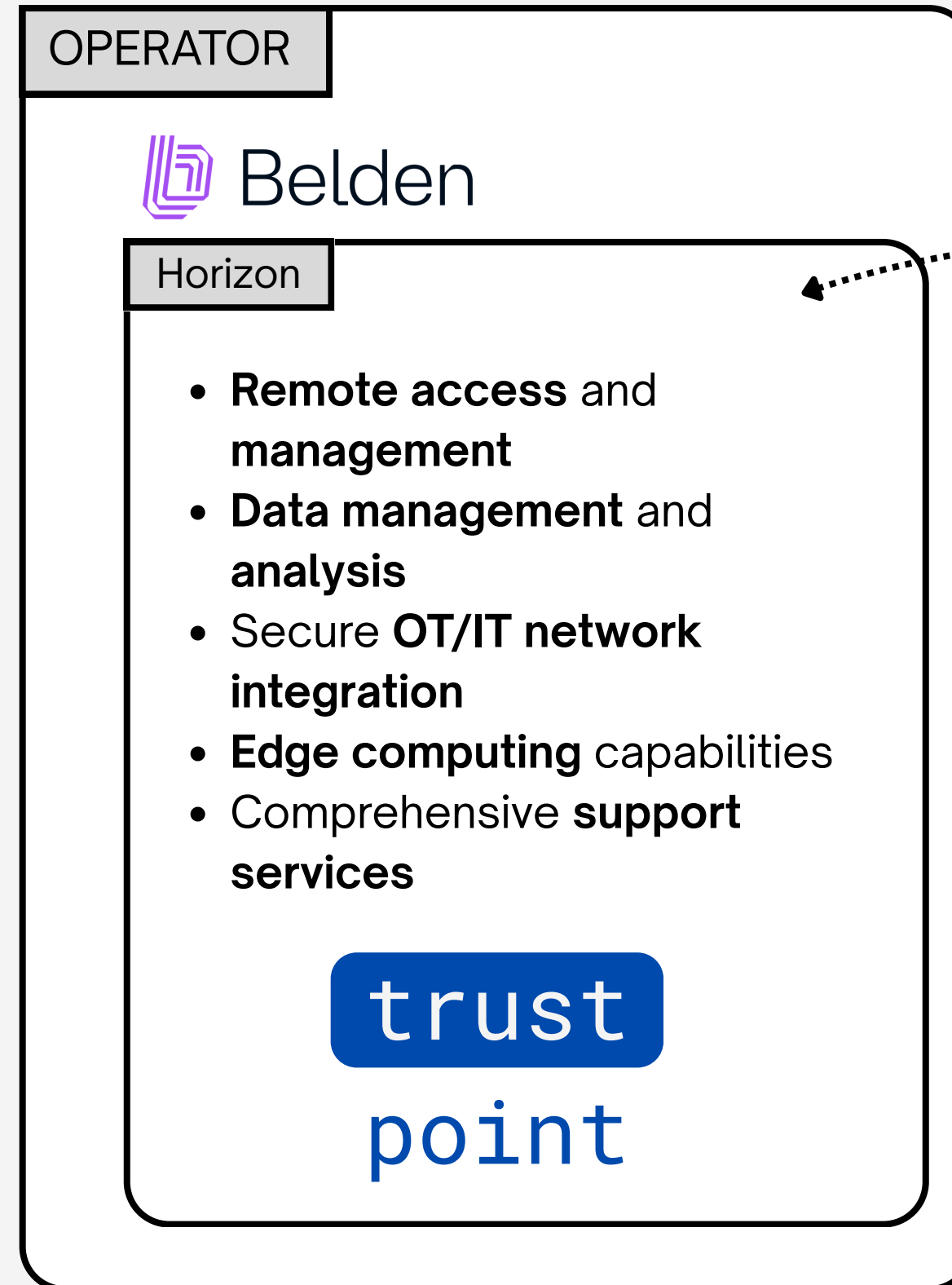
Establishing domain-level authorization and cryptographic device ownership binding to enable secure and controlled zero-touch onboarding.



PROOF OF CONCEPT

Validating secure ownership binding and zero-touch onboarding in a real industrial ecosystem

- **Integration** of Trustpoint with Belden Horizon (PoC)
- **Belden-operated** PKI issuing DevOwnerIDs
- **Operator-side** Trustpoint instance acting as domain authority
- **End-to-end onboarding** using AOKI



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More information
around **Belden Horizon**

RESSOURCES & ACCESS

Everything You Need to Explore, Deploy, and Contribute

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automation

THANK YOU.

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